

Roll No.

TCS-706

**B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

ARTIFICIAL INTELLIGENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* sub-questions among (a) and (b) in each main question.

(iii) Each question carries 10 marks.

1. (a) Explain intelligence and artificial intelligence. How do you distinguish between the two ? Write a short note on Turing Test. (CO1)

OR

- (b) What are state space search and problem formulation ? Discuss the relative merits of Depth-first and Breadth-first searching methods.

(CO3)

2. (a) Explain the concept of The Natural Language Processing System. Describe in detail the difference between natural language understanding and language generation. Explain transition Net, RTN' and ATN in NLP. (CO2)

P. T. O.

(2)

OR

- (b) What is parsing ? Parse the sentence "Tom ate an apple" using TOP : DOWN parsing using grammar. Validate sentence as per grammar :

(CO2)

- sentence → noun phrase, verb phrase
- noun phrase → proper noun
- noun phrase → determiner, noun
- verb_phrase → verb, noun_phrase
- proper noun → Tom
- noun → [apple]
- verb → [ate]
- determiner → [an]

3. (a) How to specify a task environment. Illustrate the process with a example. What do you mean by PEAS ? Determine the PEAS values for the following All problems :

(CO1)

Taxi Driver

OR

- (b) Explain AI agents design philosophy using neat and clean diagrams.

(CO2)

4. (a) Explain hill climbing and A* searching technique in detail with an example.

(CO3)

OR

- (b) What is the difference between Strong Artificial Intelligence and Weak Artificial Intelligence ? Difference between AI, ML and DL.

(CO1)

5. (a) Explain the concept and all phases of Natural Language Processing System.

(CO2)

OR

- (b) Enumerate the classical "Water Jug problem". Describe the state space for this problem. Solve this problem by giving its opera loss sequence.

(CO3)